

EventSphere: MongoDB Event Management System

Database Design

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Summary

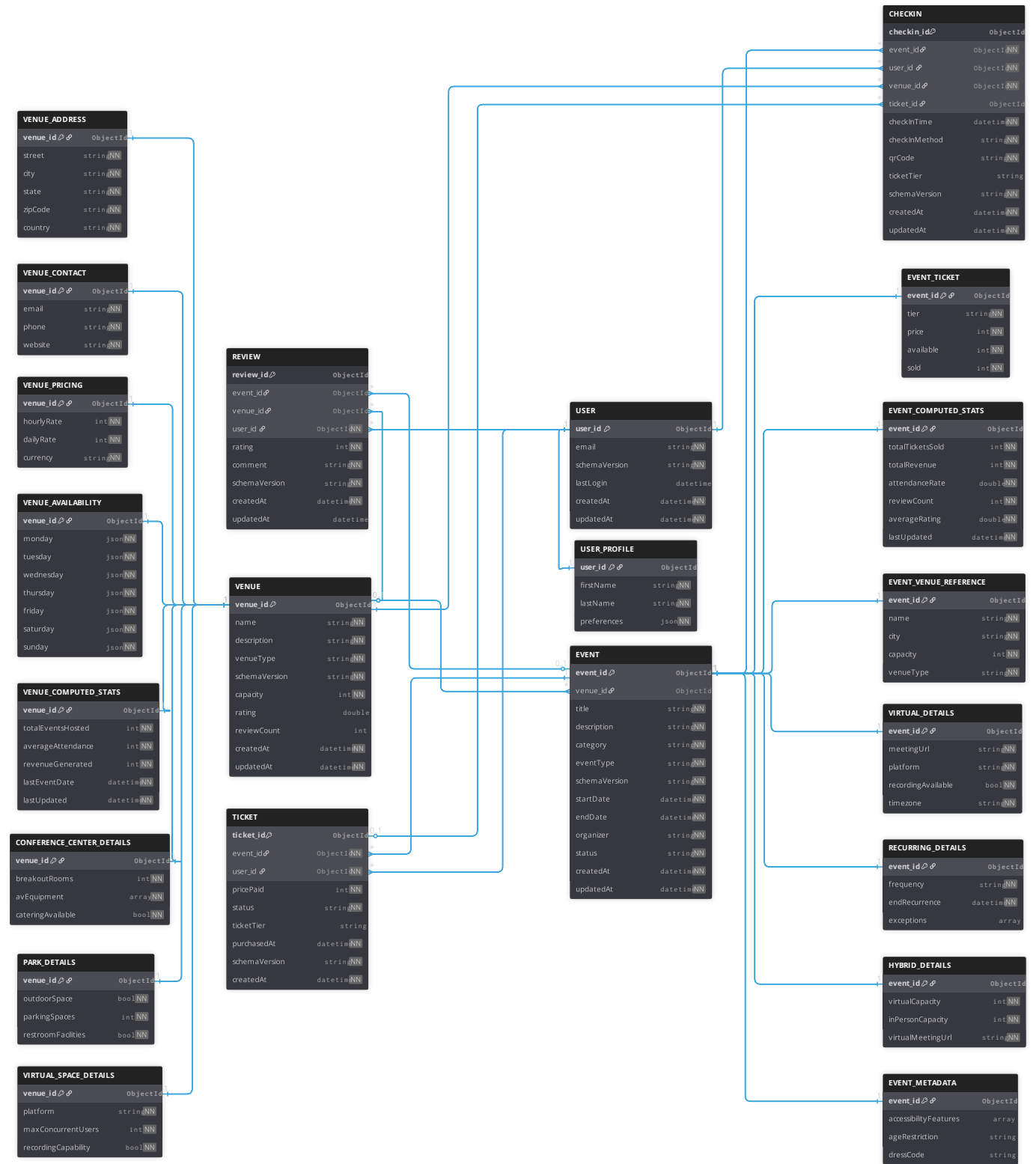
EventSphere is a MongoDB-based event management system that demonstrates advanced NoSQL database concepts through real-world application design. The system enables users to discover, review, and attend in-person, virtual, and hybrid events with geospatial discovery, full-text search, and index-optimized analytics capabilities.

This document focuses on the database design rationale, including Entity-Relationship diagram, Collection-Relationship diagram, and detailed explanations of the design patterns and architectural decisions that shaped the database structure.

Note:

- For complete schema validation rules, see `create_collections.js`.
 - For index definitions, see `create_indexes.js`.
 - For query examples, see `geospatial_aggregations.js`, `text_search_aggregations.js`, `date_range_aggregations.js`, and `analytics_aggregations.js`.
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Entity-Relationship Diagram (ERD)



ERD Design Notes

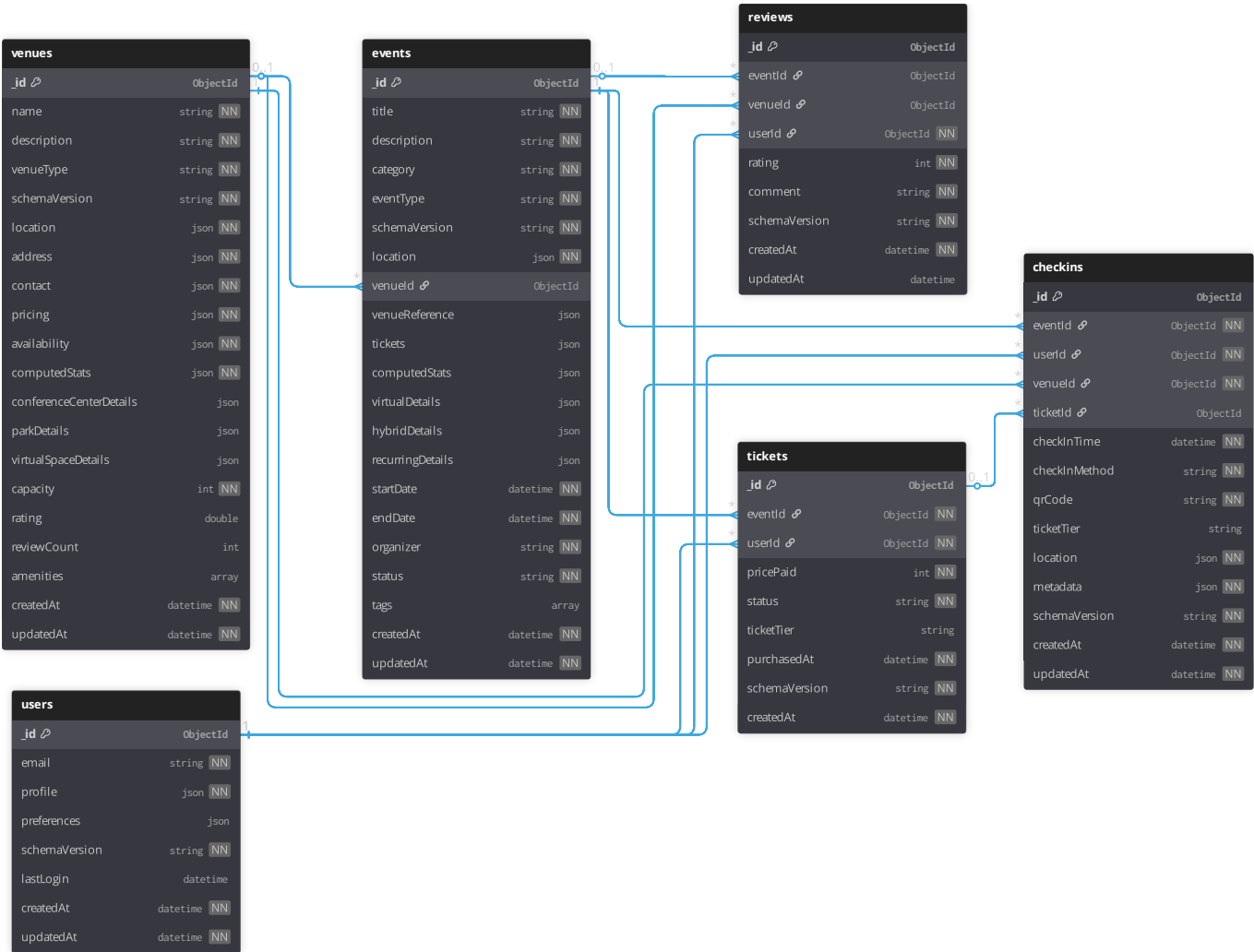
NN refers to not nullable fields.

The ERD represents the logical data model before MongoDB-specific optimizations. Key relationships:

Relationship	Type	Description
Venue → Events	1:N	One venue hosts many events
User ↔ Events	M:N	Many users attend many events (via Checkins)

Relationship	Type	Description
User → Reviews	1:N	One user writes many reviews
Event → Reviews	1:N	One event has many reviews
Venue → Reviews	1:N	One venue has many reviews
User → Tickets	1:N	One user purchases many tickets
Event → Tickets	1:N	One event has many ticket purchases
Checkin → Ticket	1:1 (optional)	A checkin may reference a purchased ticket

Collection-Relationship Diagram (CRD)



CRD Design Notes

NN refers to not nullable fields.

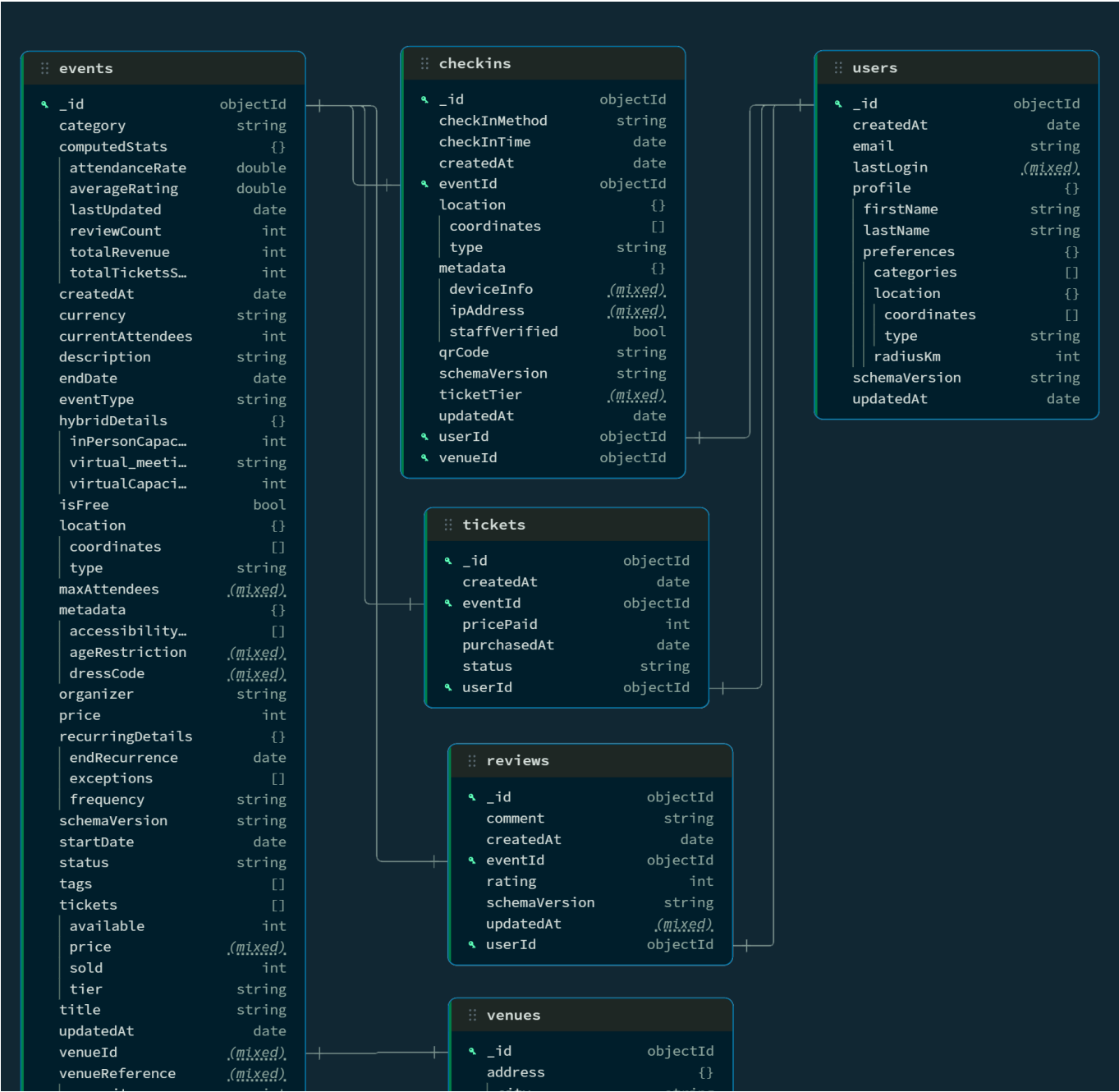
The CRD shows how the logical ERD was transformed into MongoDB collections with strategic denormalization:

Collection	Document Count	Key Embedding/Reference Decisions
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Collection	Document Count	Key Embedding/Reference Decisions
events	150,000	Embeds: tickets array, venueReference. References: venueId
venues	30,000	Embeds: address, contact, pricing, type-specific details
users	60,000	Embeds: profile with nested preferences and location
tickets	550,000	References: eventId, userId
checkins	150,241	References: eventId, userId, venueId, ticketId (optional)
reviews	149,151	References: eventId OR venueId (polymorphic), userId

MongoDB Collection Diagram Comparison

Comparison: An interesting comparison is that the MongoDB Collection Diagram is obviously similar to the CRD. But since MongoDB is reading schema from the database, it nicely displays the polymorphic design pattern in **venues** and **events**, giving a more concrete representation of the data creation.



capacity	int	city	string
city	string	country	string
name	string	state	string
venueType	string	street	string
virtualDetails	{}	zipCode	string
meetingUrl	string	amenities	[]
platform	string	availability	{}
recordingAvai...	bool	friday	{}
timezone	string	close	string
		open	string
		monday	{}
		close	string
		open	string
		saturday	{}
		close	string
		open	string
		sunday	{}
		close	string
		open	string
		thursday	{}
		close	string
		open	string
		tuesday	{}
		close	string
		open	string
		wednesday	{}
		close	string
		open	string
		capacity	int
		computedStats	{}
		averageAttend...	int
		lastEventDate	date
		lastUpdated	date
		revenueGenera...	int
		totalEventsHo...	int
		conferenceCente...	{}
		avEquipment	[]
		breakoutRooms	int
		cateringAvail...	bool
		contact	{}
		email	string
		phone	string
		website	string
		createdAt	date
		description	string
		location	{}
		coordinates	[]
		type	string
		name	string
		parkDetails	{}
		outdoorSpace	bool
		parkingSpaces	int
		restroomFacil...	bool
		pricing	{}
		currency	string
		dailyRate	int
		hourlyRate	int
		rating	double
		reviewCount	int
		schemaVersion	string
		type	string
		updatedAt	date
		venueType	string
		virtualSpaceDet...	{}
		maxConcurrent...	int
		platform	string
		recordingCapa...	bool

Advanced Design Patterns

1. Polymorphic Design Pattern

What it is: A single collection stores documents of different "types" that share a common base structure but have type-specific attributes. A discriminator field (**eventType** or **venueType**) identifies which type each document belongs to.

Why EventSphere uses it: Events and venues have diverse characteristics that would be awkward to model with separate collections or nullable fields everywhere.

Event Polymorphism

Event Type	Type-Specific Fields	Real-World Example
inPerson	Standard venue-based event	Vancouver Food Festival
virtual	virtualDetails: platform, meetingUrl, recordingAvailable, timezone	AI Workshop via Zoom
hybrid	hybridDetails: virtualCapacity, inPersonCapacity, virtualMeetingUrl	Tech Conference with live stream
recurring	recurringDetails: frequency, endRecurrence, exceptions	Monthly Book Club

Why this pattern?

- **Schema flexibility:** New event types can be added without schema migrations
- **Query efficiency:** All events live in one collection with shared indexes
- **Application simplicity:** The polymorphic field drives conditional logic in the application

Venue Polymorphism

Venue Type	Type-Specific Fields	Example
conferenceCenter	breakoutRooms, avEquipment, cateringAvailable	Vancouver Convention Centre
park	outdoorSpace, parkingSpaces, restroomFacilities	Queen Elizabeth Park
restaurant	Menu details, reservation info	Local restaurant for private events
virtualSpace	platform, maxConcurrentUsers, recordingCapability	Zoom/Teams virtual venue
stadium	Seating sections, event facilities	BC Place
theater	Stage details, seating configuration	Queen Elizabeth Theatre

Why this pattern?

- **Avoid sparse documents:** Without polymorphism, every venue would have nullable fields for all possible venue types (messy and unnecessary in MongoDB).

- **Enable type-specific queries:** "Find conference centers with AV equipment" uses `venueType` + nested field queries
- **Future extensibility:** Adding `coworkingSpace` or `warehouse` types requires no schema changes (just add a new type to the enum).

2. Extended Reference Pattern

What it is: Frequently accessed data from a referenced document is copied (denormalized) into the referencing document to avoid joins (`$lookup`) on read-heavy operations.

Why EventSphere uses it: Event listings are the most common read operation. Users need to see venue name, city, and capacity without querying the venues collection.

Implementation in Events:

```
{
  "venueId": ObjectId("..."),           // Full reference for updates etc.
  "venueReference": {                   // Extended reference for read
    optimization
    "name": "Vancouver Convention Centre",
    "city": "Vancouver",
    "capacity": 2500,
    "venueType": "conferenceCenter"
  }
}
```

Trade-offs considered:

Benefit	Cost
Event listings load in single query	Venue updates require updating all linked events (less frequent than event updates)
Enables venue-based filtering without <code>\$lookup</code>	Data duplication (acceptable for read-heavy workload like this one)
Reduces query complexity in application	Must maintain consistency on venue name/city changes (extra work for the application layer)

Why this trade-off is acceptable:

- Venue names/cities change rarely (maybe once per year)
- Event reads happen thousands of times per day
- The denormalized fields are small (4 fields, ~100 bytes)
- Application layer handles updates to both documents when venue details change

Queries enabled by this pattern:

- "Events at parks in Vancouver" - single collection query with `venueReference.venueType` and `venueReference.city`
 - "Events at conference centers with capacity > 500" - compound query on embedded fields
-

3. Computed Pattern

What it is: Pre-calculated statistics are stored directly in documents to avoid expensive aggregation queries on every read.

Why EventSphere uses it: A dashboard or analytics page for an Event organizer needs to show ticket sales, revenue, ratings, and attendance rates. Computing these on every page load would require aggregating across multiple collections.

Event Computed Statistics:

```
"computedStats": {
  "totalTicketsSold": 125,
  "totalRevenue": 16875,
  "attendanceRate": 25.0,
  "reviewCount": 8,
  "averageRating": 4.3,
  "lastUpdated": ISODate("2025-10-01T23:16:16.047Z")
}
```

Venue Computed Statistics:

```
"computedStats": {
  "totalEventsHosted": 156,
  "averageAttendance": 850,
  "revenueGenerated": 2450000,
  "lastEventDate": ISODate("2025-09-28T23:16:15.999Z"),
  "lastUpdated": ISODate("2025-10-01T23:16:15.999Z")
}
```

Update strategy:

- `lastUpdated` field tracks when stats were last recalculated
- Stats can be updated via scheduled background jobs or triggered on write operations (cron job or webhook)
- For demo/academic purposes, stats are pre-computed during data generation

Why this pattern?

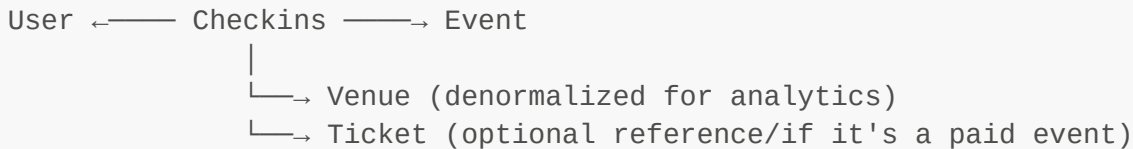
- **Performance:** This would drop the load time of a dashboard or analytics page drastically.
 - **Scalability:** Aggregation cost is spread across writes, not multiplied across reads
 - **Single source of truth:** One place to look for event/venue metrics
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4. Bridge Collection Pattern (Checkins)

What it is: A dedicated collection that represents a many-to-many relationship, storing the relationship itself along with relationship-specific attributes.

Why EventSphere uses it: Users attend many events, events have many attendees. Embedding attendees in events would cause document bloat (1000+ attendee arrays). Embedding events in users has the same problem.

Checkins as a Bridge Collection:



Why not embed attendees in events?

- MongoDB documents have a 16MB limit
- Large events (NFL games, concerts) could have 50,000+ attendees
- Each attendee addition would rewrite the entire event document
- Attendee queries ("what events has user X attended?") would require scanning all events

Why not embed events in users?

- Active users might attend 100+ events per year
- Each event attendance would grow the user document
- User profile queries don't always need attendance history

Benefits of bridge collection:

- **Unbounded scaling:** Millions of checkins without affecting event/user document sizes
- **Relationship attributes:** `checkInTime`, `checkInMethod`, `ticketTier`, `qrCode` belong to the relationship, not the entities
- **Flexible analytics:** Easy to answer "peak check-in times", "check-ins by method", "attendance patterns"
- **Unique constraint:** Index on `{eventId: 1, userId: 1}` prevents duplicate check-ins

5. Dual Ticket Architecture

What it is: Two complementary data structures for tickets - embedded ticket types in events, and a separate collection for individual purchases.

Why EventSphere uses it: Ticket types (tiers/pricing) are always displayed with events and are bounded (small number of types). Ticket purchases are unbounded and need independent queries so they are stored in a separate collection.

Embedded EventTickets (in events collection):

```
"tickets": [
  { "tier": "Early Bird", "price": 35, "available": 500, "sold": 250 },
  { "tier": "General Admission", "price": 45, "available": 1500, "sold": 800 },
  { "tier": "VIP", "price": 150, "available": 50, "sold": 45 }
]
```

Separate Tickets Collection (user purchases):

```
{
  "eventId": ObjectId("..."),
  "userId": ObjectId("..."),
  "ticketTier": "VIP",
  "pricePaid": 150.00,
  "status": "active",
  "purchasedAt": ISODate("...")
}
```

Why this architecture?

Aspect	Embedded (ticket types)	Separate (purchases)
Size	Bounded (1-5 items)	Unbounded (millions)
Access pattern	Always with event	Independent queries
Example query	"Show event with pricing"	"User's purchased tickets"
Update frequency	Rarely (pricing changes)	Frequently (each purchase)

Industry precedent: I read that Ticketmaster, Eventbrite, and StubHub use similar patterns - catalog data embedded, and transaction data separate.

6. Checkin-Ticket Relationship Pattern

What it is: Checkins optionally reference tickets, supporting both paid and free event attendance while avoiding data duplication.

Why EventSphere uses it: Not all check-ins require tickets. Free events, walk-ins, staff, and volunteers check in without purchasing.

Pattern implementation:

- **70% of checkins:** Have `ticketId` linking to purchased ticket
- **30% of checkins:** `ticketId` is null (free events, walk-ins)
- **Denormalized field:** `ticketTier` kept in checkin for quick display

Why denormalize ticketTier?

- Displaying check-in lists shouldn't require \$lookup to tickets
- Tier doesn't change after purchase (safe to denormalize)
- Only one field duplicated (minimal storage cost)

Queries enabled:

- "Which ticket was used for this checkin?" - direct lookup via `ticketId`
- "All checkins for this ticket" - query by `ticketId`
- "Mark ticket as used on checkin" - update ticket status atomically

Embedding vs. Referencing Decisions

A critical NoSQL design choice is when to embed data vs. when to reference it.

How I made this decision for EventSphere:

Embedded (denormalized)

Data	Embedded In	Rationale
Ticket types	events.tickets	Bounded (1-5), always accessed with event, rarely changes
Venue reference	events.venueReference	High read frequency, small size, enables filtering
Address	venues.address	1:1 relationship, always accessed together
Contact info	venues.contact	1:1 relationship, small size
User preferences	users.profile.preferences	1:1 relationship, user-specific
Computed stats	events/venues	Avoids expensive aggregations

Referenced (normalized)

Data	Stored In	Referenced By	Rationale
Ticket purchases	tickets	eventId, userId	Unbounded, needs independent queries
Check-ins	checkins	eventId, userId, venueId	M:N relationship, relationship attributes
Reviews	reviews	eventId OR venueId, userId	Unbounded per entity, independent queries
Full venue data	venues	events.venueId	Updates should happen in one place
User data	users	checkins.userId, reviews.userId	Single source of truth

Indexing Strategy

Strategic indexes are directly influential on query performance. EventSphere uses **24 indexes** (4 per collection) designed for common query patterns.

Full index definitions: See `create_indexes.js`

Index Design Rationale

Collection	Index	Query Pattern Supported
events	<code>location: "2dsphere"</code>	Geospatial discovery ("events near me")
events	<code>{title, description, category, tags}: "text"</code>	Full-text search with relevance
events	<code>{category: 1, startDate: 1}</code>	"Technology events this weekend"
events	<code>{eventType: 1, startDate: 1}</code>	"Virtual events next month"
venues	<code>location: "2dsphere"</code>	Venue discovery for event creation
venues	<code>{venueType: 1, capacity: 1}</code>	"Conference centers ≥500 capacity"
checkins	<code>{eventId: 1, userId: 1}</code> (unique)	Prevent duplicate check-ins
users	<code>email</code> (unique)	Authentication lookups

Index Trade-offs

- **Write performance:** Each index adds a storage overhead to the database.
- **Read performance:** Each index improves query performance.

Data Validation

All collections enforce JSON Schema validation at the database level. Key validation rules:

Validation Type	Example	Purpose
Coordinate bounds	Longitude: -180 to 180	Prevent invalid geospatial data
Required fields	title, category, location for <code>events</code>	Data integrity
Enum validation	<code>eventType</code> : ["inPerson", "virtual", "hybrid", "recurring"]	Type safety
Range validation	<code>rating</code> : 1-5, <code>price</code> : ≥0	Business rule enforcement
Unique indexes	<code>users.email</code> , <code>checkins.{eventId, userId}</code>	Prevent duplicates

Full validation schemas: See `database/schemas/create_collections.js`

Schema Versioning

All collections include a `schemaVersion` field to support future schema evolution:

- **Current version:** "1.0" for all collections
 - **Migration strategy:** Application can handle multiple versions during transitions
 - **Backward compatibility:** Old documents remain readable while new fields are added
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Conclusion

The EventSphere database design demonstrates MongoDB expertise through:

- **Strategic embedding:** Denormalizing for read performance where appropriate
- **Reference patterns:** Normalizing for scalability and independent queries
- **Polymorphic design:** Flexible schemas for diverse entity types
- **Computed patterns:** Pre-calculated stats for dashboard performance
- **Bridge collections:** Scalable many-to-many relationships with relationship attributes

These patterns reflect a well thought out and applicable NoSQL design. Similar patterns are used by industry leaders like Airbnb (geospatial), and Eventbrite (event management).